



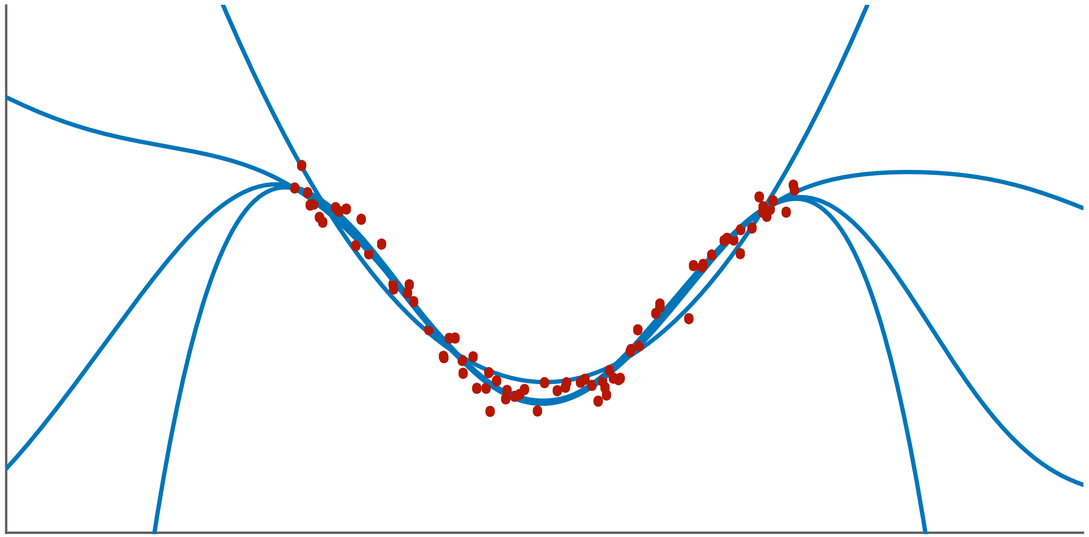
Kernel Autoresearch for Open-Ended Model Discovery

Richard Cornelius Suwandi, Feng Yin, Kevin Murphy



Cohere Labs, Open Science Community

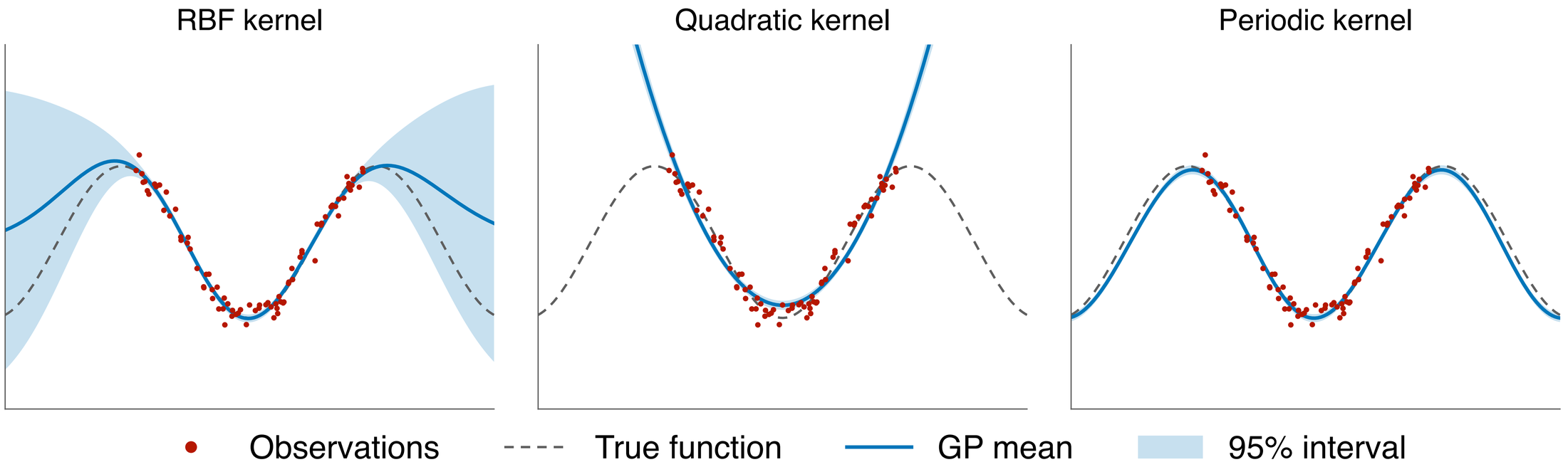
Which function should we prefer?



- Suppose we observe some **noisy samples** of an unknown function
- There are many possible functions that can fit these points
- **Inductive bias** determines which of these functions the model prefers

How can we encode **inductive bias** in a model?

Kernels as inductive biases



- Different kernels encode different **inductive biases**
- A misspecified kernel yields **biased** predictions and **miscalibrated** uncertainty

How should we choose a kernel given the data?

Compose kernels from a fixed library

Structure Discovery in Nonparametric Regression through Compositional Kernel Search

[Duvenaud et al., 2013]

"built compositionally by adding and multiplying a small number of base kernels"

Method: Search over a grammar of kernel sums and products using a model-selection score

Compose kernels from a fixed library

Structure Discovery in Nonparametric Regression through Compositional Kernel Search

[Duvenaud et al., 2013]

Differentiable Compositional Kernel Learning for Gaussian Processes

[Sun et al., 2018]

"each unit of the network corresponds to a valid kernel"

Method: Optimize differentiable kernel compositions with gradient-based training

Compose kernels from a fixed library

Structure Discovery in Nonparametric Regression through Compositional Kernel Search

[Duvenaud et al., 2013]

Differentiable Compositional Kernel Learning for Gaussian Processes

[Sun et al., 2018]

Adaptive Kernel Design for Bayesian Optimization Is a Piece of CAKE with LLMs

[Suwandi et al., 2025]

"LLMs as the crossover and mutation operators"

Method: Evolve kernel compositions using LLMs and in-context learning as data arrive

These methods search over combinations, but only using a **fixed set of base kernels**

Learn flexible kernels from data

Gaussian Process Kernels for Pattern Discovery and Extrapolation

[Wilson and Adams, 2013]

"model a spectral density […] with a Gaussian mixture"

Method: Learn a mixture of frequency components to represent a stationary kernel

Learn flexible kernels from data

Gaussian Process Kernels for Pattern Discovery and Extrapolation

[Wilson and Adams, 2013]

Deep Kernel Learning

[Wilson et al., 2016]

"we transform the inputs of a spectral mixture base kernel with a deep architecture"

Method: Jointly learn neural features and kernel parameters through the GP marginal likelihood

Learn flexible kernels from data

Gaussian Process Kernels for Pattern Discovery and Extrapolation

[Wilson and Adams, 2013]

Deep Kernel Learning

[Wilson et al., 2016]

Few-Shot Bayesian Optimization with Deep Kernel Surrogates

[Wistuba and Grabocka, 2021]

"train a shared deep surrogate model to quickly adapt (with few response evaluations)"

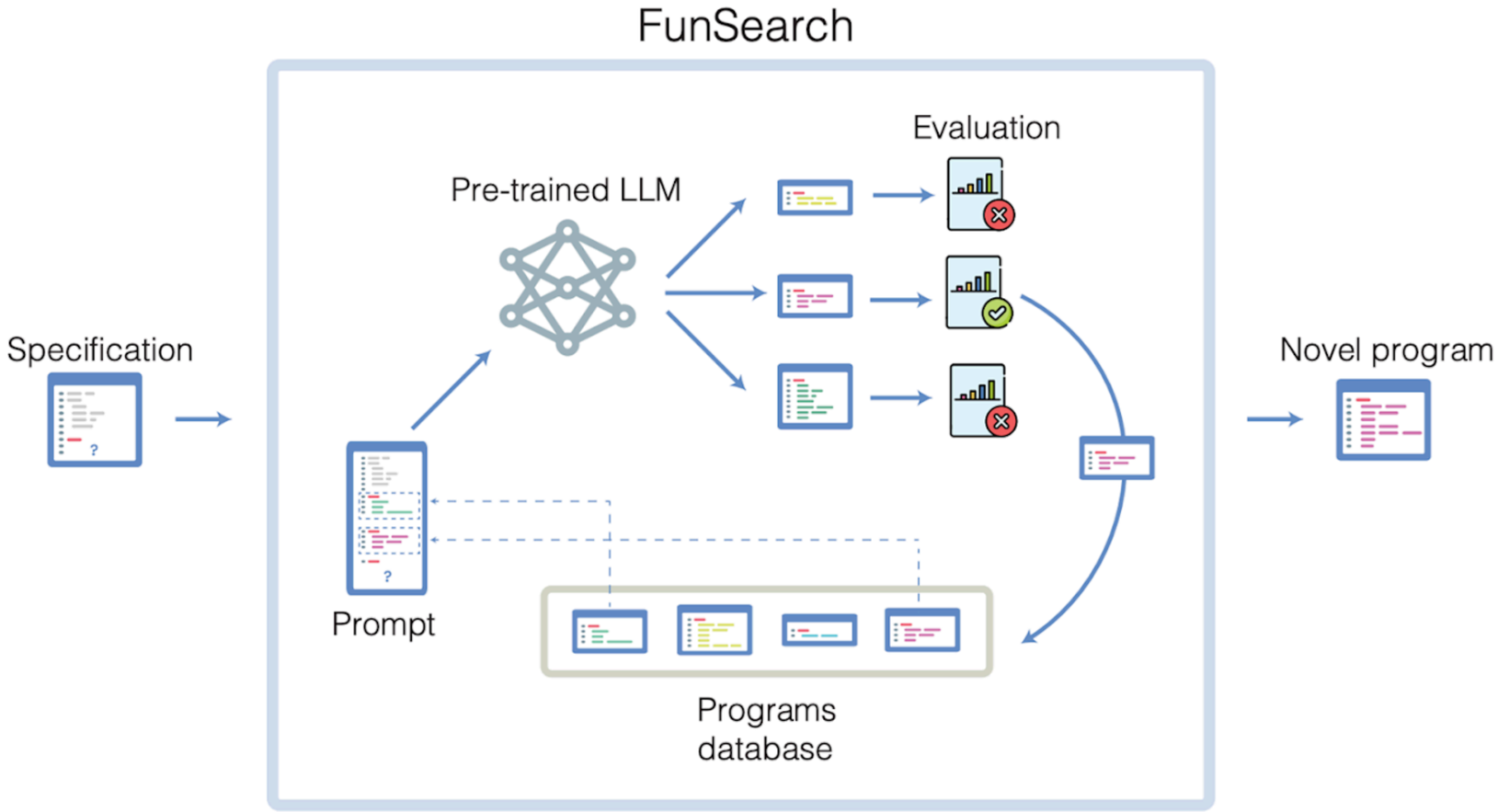
Method: Meta-learn a deep kernel across tasks, then adapt it to a new task

These methods learn kernels from data, but only within a **fixed family or architecture**

Coding agents for algorithm discovery

Recent works have shown that coding agents can discover **new algorithms**

[Romera-Paredes et al, 2024; Novikov et al., 2025]



Romera-Paredes et al., "Mathematical discoveries from program search with large language models", Nature, 2024

Kernels as programs

Concurrent work uses LLMs to write **kernels as programs**

Automated Kernel Discovery Towards Understanding High-dimensional Bayesian Optimization

Yun et al., 2026

"searches a broader kernel space beyond predefined composition rules"

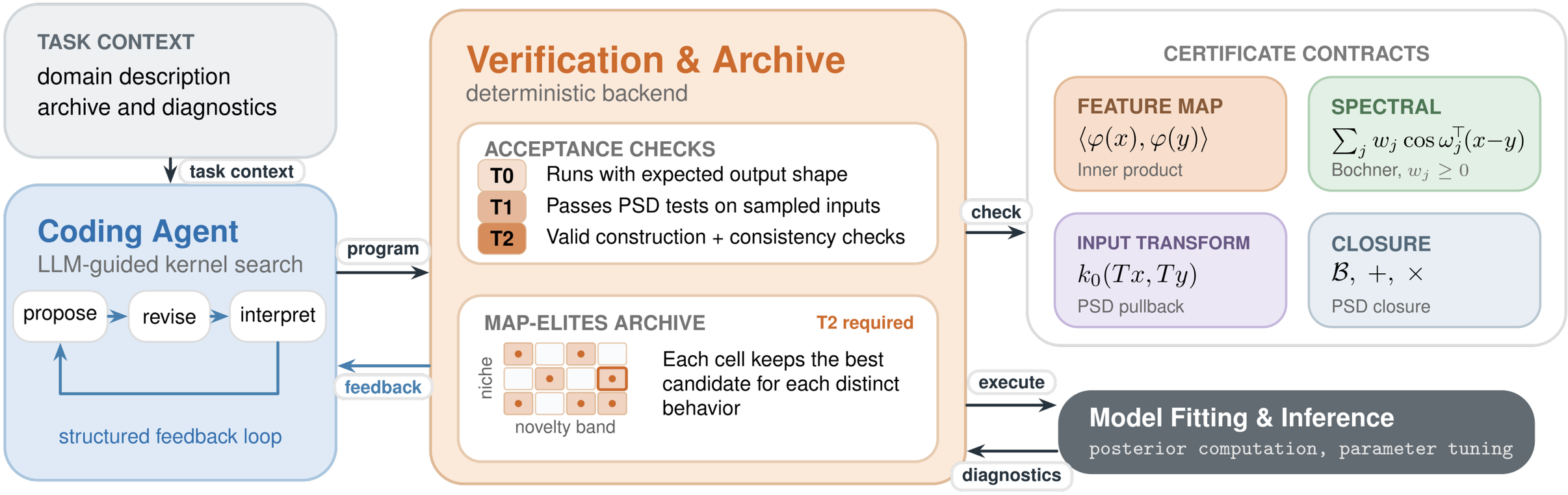
Method: Generate kernel program and screen the sampled Gram matrices with a Cholesky factorization.

- Kernel validity is undecidable for arbitrary programs [Rice, 1953]
- Numerical checks on **sampled** inputs cannot establish validity over the **entire** input domain
- In our stress tests, **22–58%** of LLM-written kernels that pass sampled checks fail at other scales or dims

Can we search **beyond** a fixed grammar and still **guarantee** validity?

Rice, "Classes of recursively enumerable sets and their decision problems", Transactions of the American Mathematical Society, 1953

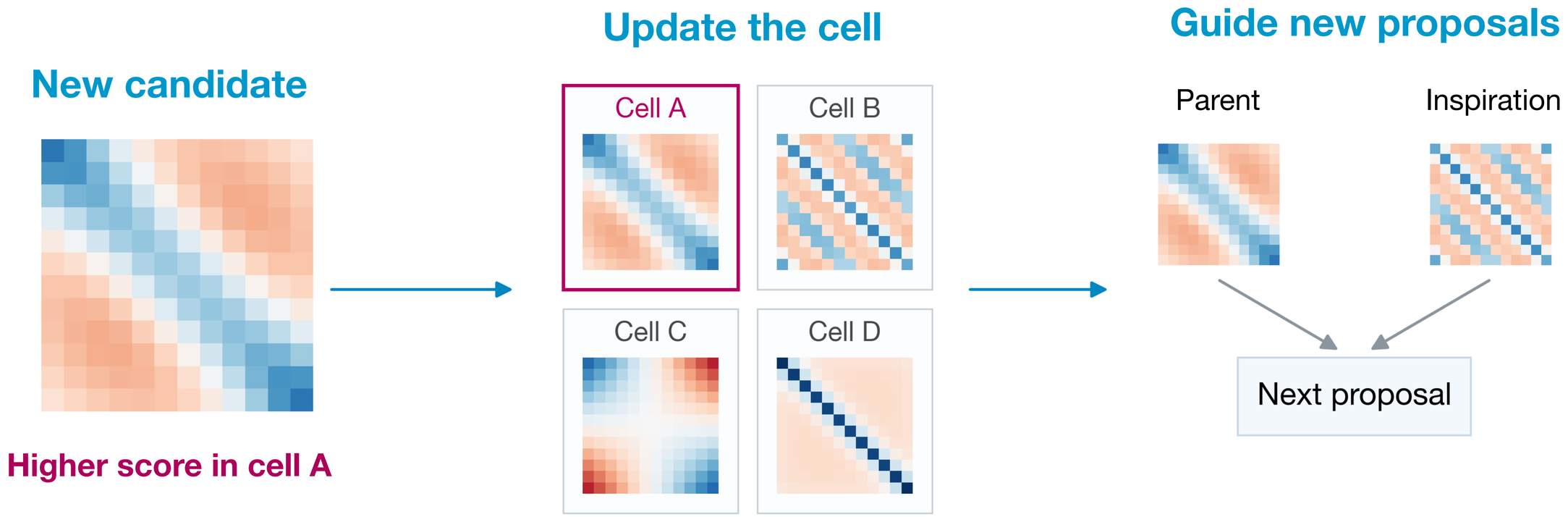
Kernel Autoresearch (Kernaut 👩‍🚀)



Suwandi et al., "Kernel autoresearch for open-ended model discovery", arXiv, 2026

Quality-diversity archives

We use MAP-Elites to keep **diverse, strong** kernels during search [Mouret & Clune, 2015]



Mouret and Clune, "Illuminating search spaces by mapping elites", arXiv, 2015

Novelty screening

- The agent may rediscover a "known" kernel under a new name/notation/implementation
- To detect this, we compare their **kernel fingerprints**:

$$d_{\text{nov}}(\kappa) = \min_{r \in \mathscr{R}} \frac{1}{|\mathscr{D}|} \sum_{d \in \mathscr{D}} \left\| \bar{K}_{\kappa}(X_d) - \bar{K}_{r}(X_d) \right\|_F$$

nearest known kernel

average over probe dimensions

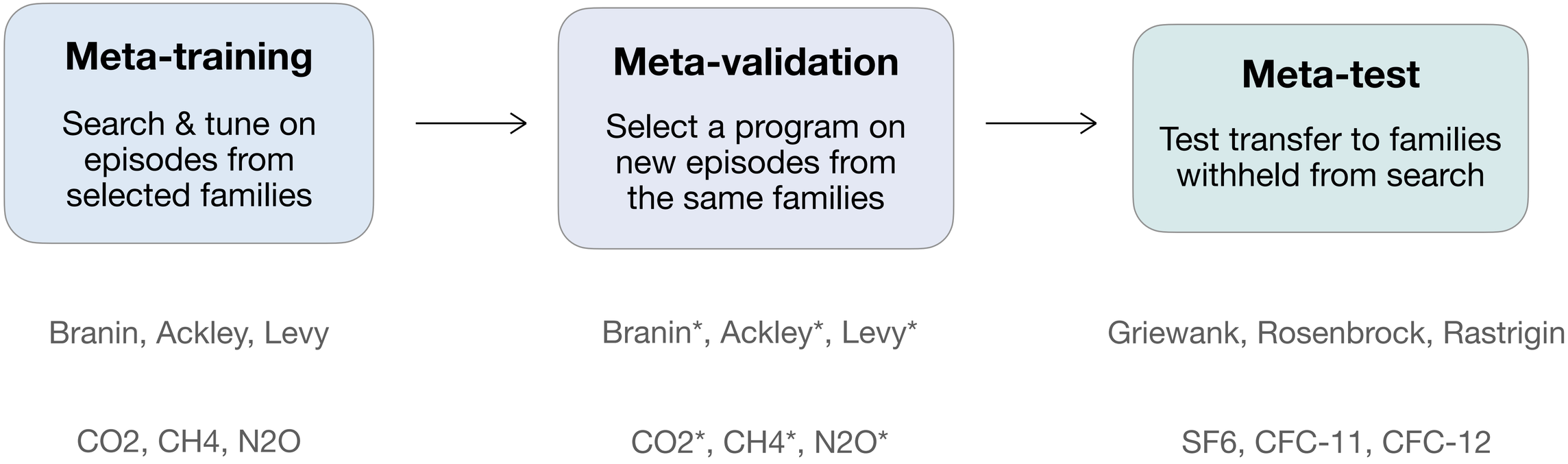
candidate fingerprint

known fingerprint

- The **reference set** includes standard kernels and kernels discovered earlier in the search

Meta-evaluation

- We seek kernel programs that **transfer to unseen tasks** without repeating discovery
- We adopt a **meta-evaluation** framework inspired by [Goldie et al., 2026]



Goldie et al., "DiscoGen: Procedural Generation of Algorithm Discovery Tasks in Machine Learning", ICML, 2026

Evaluation metrics

- We score predictions with the **continuous ranked probability score (CRPS)**:

$$\mathrm{CRPS}\big(\mathcal{N}(\mu, \sigma^2), y\big) = \sigma\left[z\big(2\Phi(z) - 1\big) + 2\phi(z) - \frac{1}{\sqrt{\pi}}\right], \qquad z = \frac{y - \mu}{\sigma}$$

GP predictive distribution

observed value

standardized error

- CRPS evaluates the full predictive distribution, so lower values mean **more accurate and better-calibrated predictions** [Gneiting and Raftery, 2007]

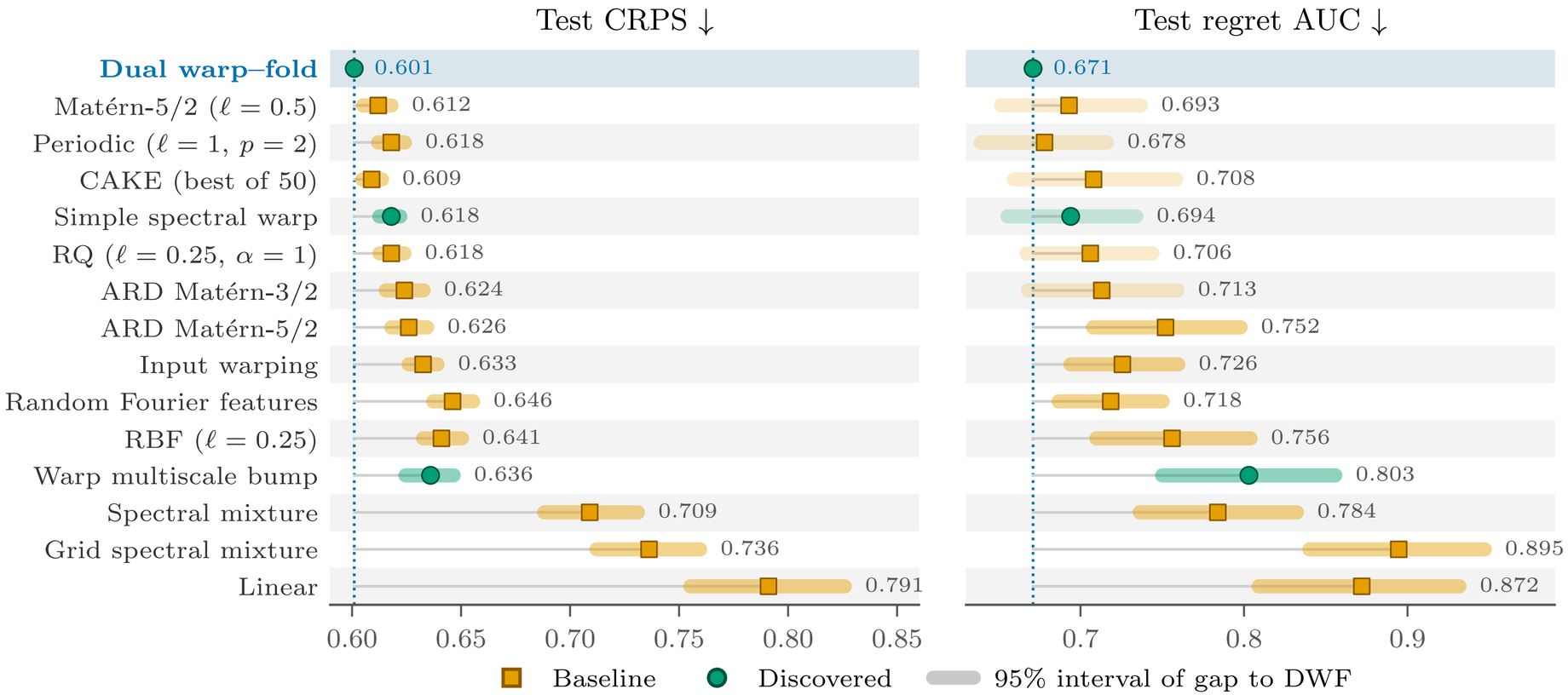
- For black-box optimization, we also report the **regret AUC**:

$$\mathrm{RegretAUC} = \frac{1}{T}\sum_{t=1}^{T}\frac{f^\star - y_t^\star}{f^\star - y_0^\star}$$

average over T steps

gap to the optimum after step t, relative to the initial gap

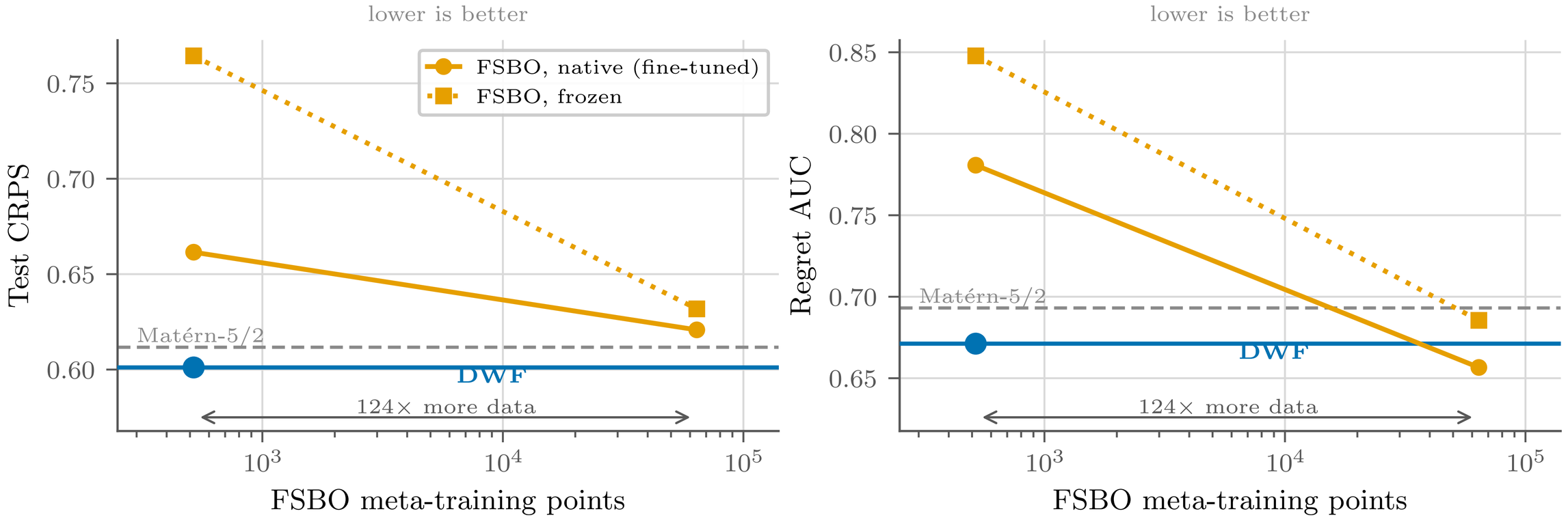
Transfer to unseen black-box optimization functions



- Test tasks come from six function families withheld from search, with coordinates randomly permuted, reflected, and warped in every episode

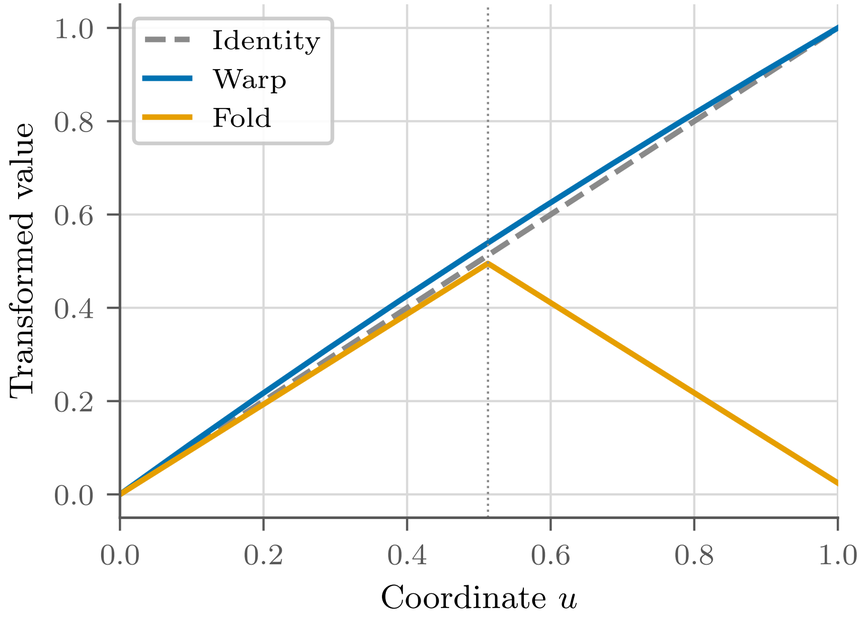
- The discovered kernel, **dual warp–fold (DWF)**, achieves the lowest test error and regret among all fixed and learned kernels

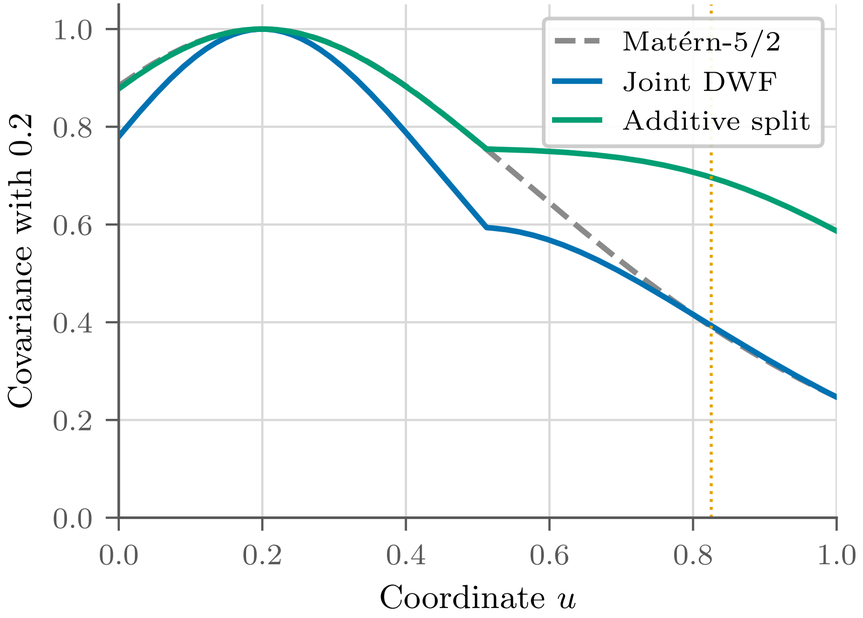
Transfer to unseen black-box optimization functions



- With the **same meta-training points**, DWF beats FSBO, a meta-learned deep neural kernel, on both predictive error and regret, while FSBO needs about **124× more data** to approach it
- A short program with the appropriate structure can therefore be substantially **more data-efficient** than a learned neural representation

Discovered kernel: Dual-Warp Fold (DWF)



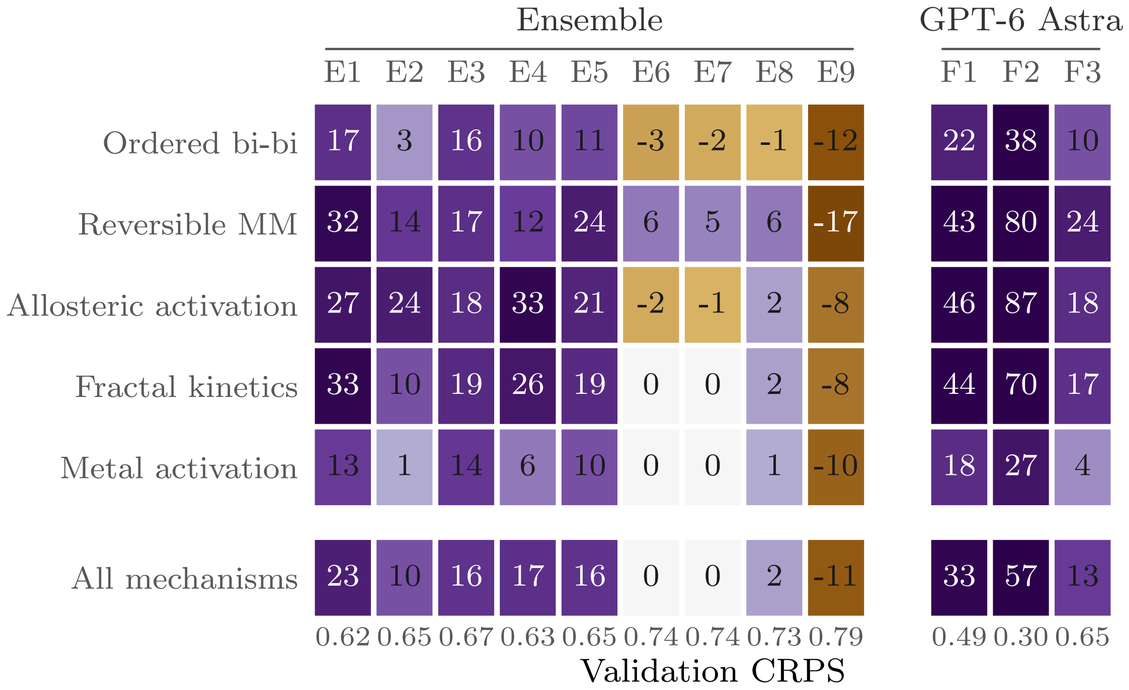


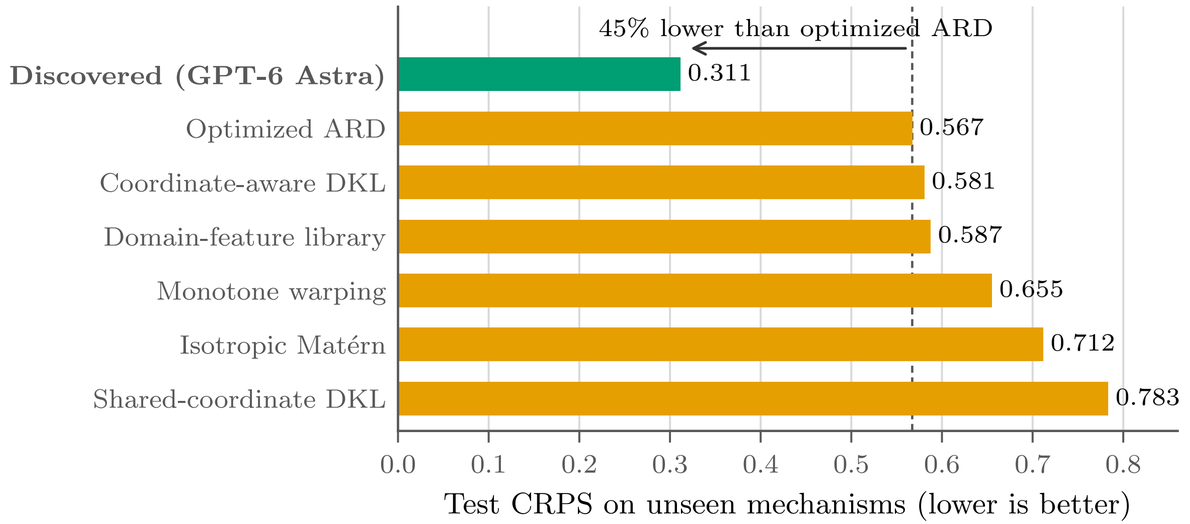
$$k(\mathbf{x}, \mathbf{x}') = \underbrace{k_{0.5}^{5/2}}_{\text{Matérn base}}\left(\underbrace{T(\mathbf{x}), T(\mathbf{x}')}_{\text{discovered coordinates}}\right) \qquad T(\mathbf{x}) = \Bigg(\underbrace{\frac{1 - e^{-\lambda x_j}}{1 - e^{-\lambda}}}_{\text{gentle monotone warp}} , \underbrace{\frac{\sin \zeta}{\pi} \arcsin\Big(\sin(\pi \nu x_j)\Big)}_{\text{triangular fold}} \Bigg)_{j=1}^{d}$$

- Stationary-kernel sums and products **cannot express** DWF's location-dependent similarity
- A human-designed refinement of this structure **further reduced** held-out CRPS by 5.7%

Transfer to new enzyme mechanisms

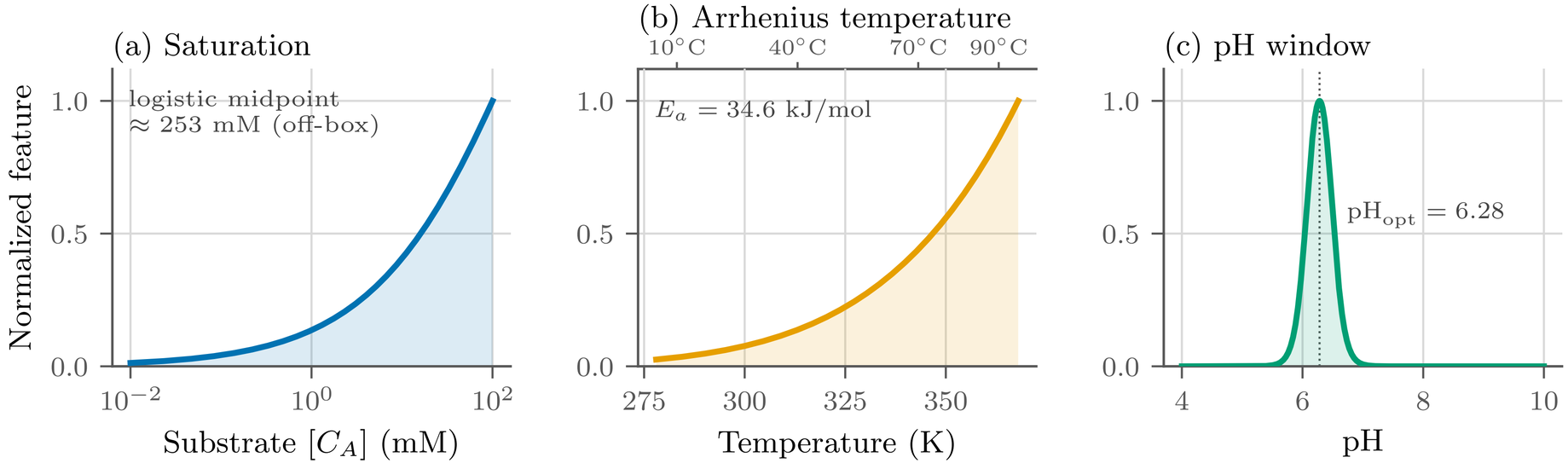
(a) Enzyme kinetics: CRPS reduction vs. Matérn-5/2 (%)





- Kernels discovered on **ten canonical rate laws** predict reaction rates for **five unseen enzyme mechanisms**
- The discovered kernel (GPT-6 Astra) **reduces error by 45%** relative to optimized ARD and beats all six learned baselines, including deep kernel learning (DKL)

Discovered kernel: enzyme kinetics



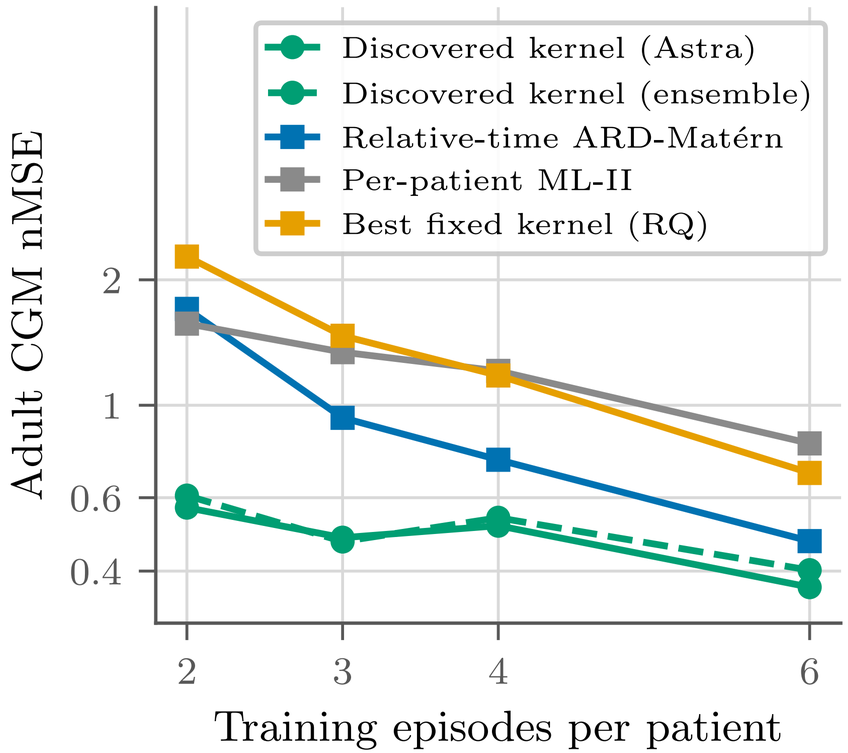
$$k(\mathbf{x}, \mathbf{x}') = k_{\mathrm{RBF}} \left(T(\mathbf{x}), T(\mathbf{x}') \right)$$

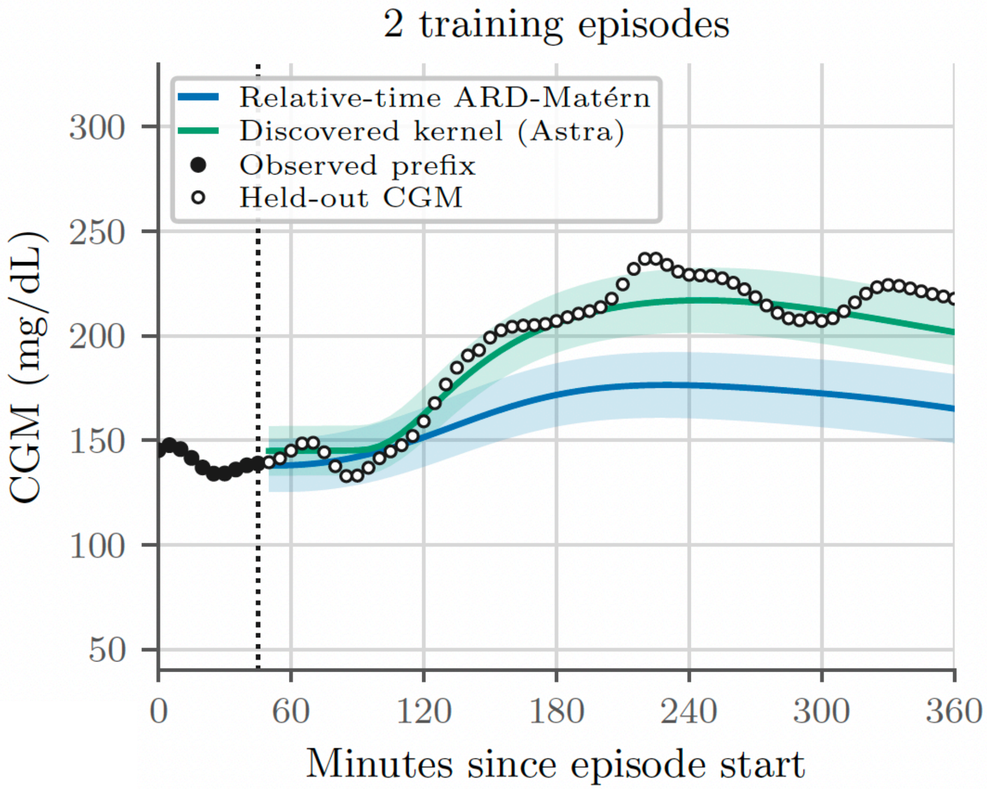
base kernel discovered features

$$T(\mathbf{x}) = \left(\xi_A s(a),\ \xi_A s(b),\ e,\ \xi_I i\, s(a),\ \frac{1}{2} p\, s(a),\ \xi_T e^{-E_a/(RT_K(t))},\ \xi_{\mathrm{pH}} q(h),\ \frac{1}{2} s(a) s(b),\ \frac{1}{2} \right)$$

substrate responses temperature pH window interaction

Transfer to new glucose patients

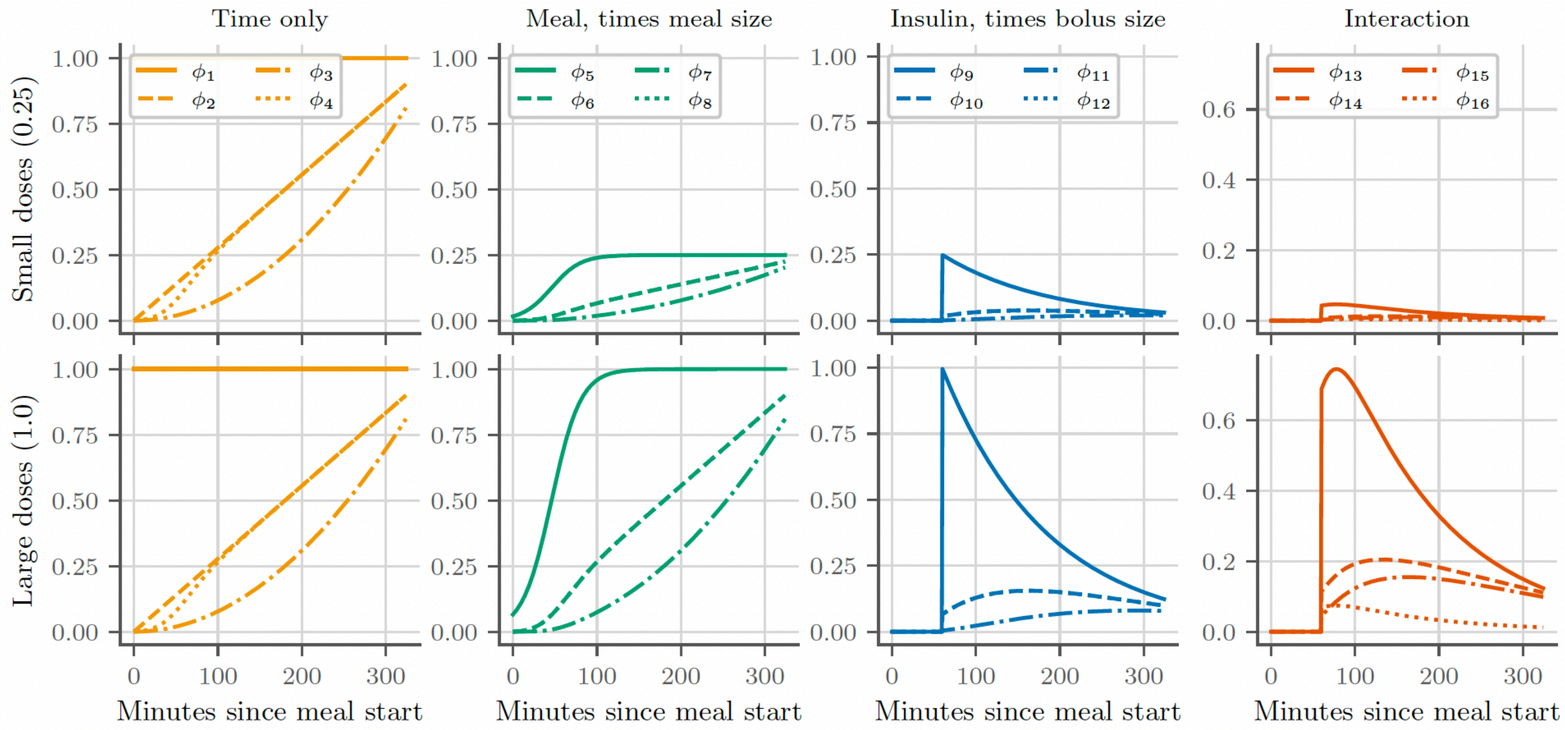




- Kernels discovered on children and selected on adolescents **forecast blood glucose** for **unseen adult patients** after a 45-minute observed prefix

- With only two training episodes per patient, the discovered kernels **reduce error by 38–67%** relative to the strongest tuned baseline

Discovered kernel: meal and insulin responses



$$k(\mathbf{x}, \mathbf{x}') = \underbrace{\varphi(\mathbf{x})^\top \varphi(\mathbf{x}')}_{\text{inner product of 16 features}}$$

$$\tau_m = t - t_{\text{meal}}$$
$$\tau_b = t - t_{\text{bolus}}$$
$$q(\tau) = (1, \tau, \tau^2, |\tau|)$$
$$m, b = \text{normalized doses}$$

$$\varphi(\mathbf{x}) = \Big(\underbrace{1, \tau_m, \tau_m^2, g_m \tau_m}_{\text{time features}}, \underbrace{m g_m q(\tau_m)}_{\text{logistic meal response}}, \underbrace{b g_b q(\tau_b)}_{\text{delayed insulin decay}}, \underbrace{m b g_m g_b (1, \tau_m, \tau_b, |\tau_m - \tau_b|)}_{\text{meal–insulin interaction}} \Big)$$

- The discovered kernel computes the time since the meal and since the insulin bolus, and builds **16 explicit response features** from them

Takeaways

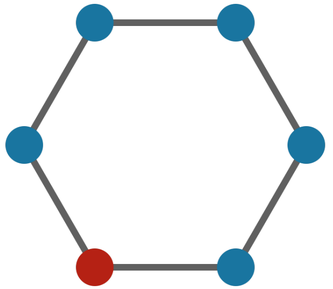
1. Coding agents can discover **reusable inductive biases** as kernel programs
2. We can let the agent propose **kernel components** freely while **construction contracts** guarantee that every accepted kernel is valid
3. A **quality-diversity archive** and **novelty screening** can steer the agent toward diverse, functionally new kernels
4. We can **amortize the cost of discovery**, because the discovered kernel can be reused on new tasks without any further calls to the agent
5. Open-ended search over kernel programs **outperforms hand-designed** kernel grammars and architectures

Ongoing and Future Work

Extend the construction contracts to **new kinds of kernels**

- **Kernels for structured inputs** such as strings and graphs
- **State-space kernels** derived from linear ODEs
- **Semantic kernels** for LLM uncertainty

A C G T T G C A

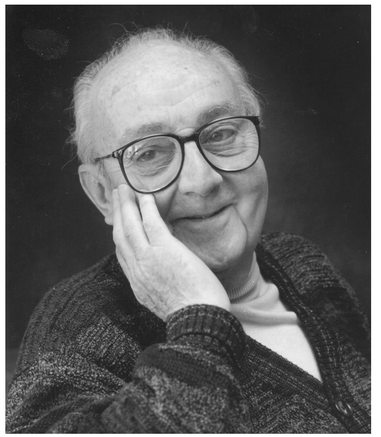


We welcome **contributions** of new tasks, domains, and construction contracts!

Thank you!

George E.P. Box (1979):

“All models are wrong,
but some are **useful**.”



arxiv.org/abs/2610.10394

github.com/richardcsuwandi/kernaut

richardsuwandi@link.cuhk.edu.cn

Scan for the paper